On The Gravitational Nature Of Time

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Abstract

We explore a modified set of coordinate transformations that describe time, cosmic inflation and gravitational acceleration as the same process as observed from different reference frames through an unwrapped Kaluza-Klein like geometry. I'LL HAVE AI DO THIS WHEN THE PAPER'S DONE.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of the geometry of SR under the assumption that the $\mathbb{R}^3$ space between two arbitrary points dilates as time progresses, motivated by the position that classical simultaneity is a fundamental property of our reality.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6]. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time where the 4th KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th axis.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event exists at a single point in space-time; a local measurement as $dt \to 0$. The constant nature of 4-vecs inherent to the definition of an event in SR arises naturally from the coordinate transformations that follow, giving d/v that is equivalent as measured in each respective coordinate system.

Let there be two overlapping coordinate systems, **S** and **U**. Let **S** and **U** represent static positions in time, with **U** representing a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{1}$$

where $\Delta t > 0$.

Let there also be two observers, A and B. Let A and B agree that B should travel at some relativistic velocity v' between two arbitrary points in space contained within **S** and **U**, p_1 and p_2. Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

If A and B were to disagree upon the time at which B reaches p_2, this single, instantaneous event (B reaching p_2) would be required to happen more than once. Therefore, we theorize that it is not possible for A and B to disagree upon the time at which B reaches p_2.

Instead, as the additional factor of γ has been demonstrated experimentally on a number of occasions[7–11], we should conclude that both A and B observe B reaching p_2 in $\frac{d}{v}\gamma$. This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \frac{\overbrace{d \cdot \gamma}^{B}}{v^{(U)}} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{2}$$

This produces a model of relativistic dilation that loosely binds the additional γ factor in the $d = vt\gamma$ equivalence to cosmic inflation, where the additional velocity experienced by B results in a more rapid progression along this concurrent density axis leading to a further separation between any two points in **U**.

2 Theoretical Framework

Let us acknowledge the two primary properties of time:

1. A driving rate of change, as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance, and acts only as a mathematical tool: $t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, now denoted y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart.

As we are removing time as a physically meaningful quantity, let us define

$$\tau = \frac{\dot{x}}{\widehat{v}} \tag{3}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change. Let $\widehat{v} = v \cdot 1t$, or the velocity in units of distance.

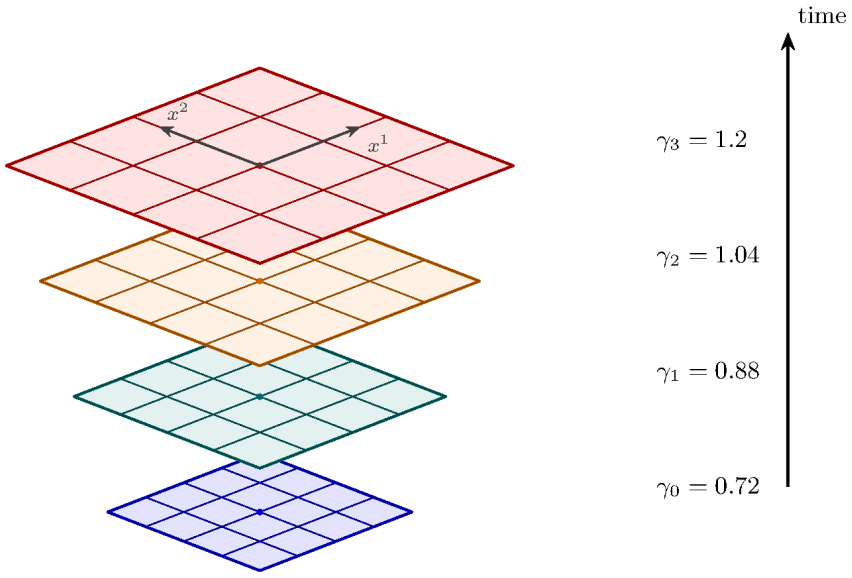


Fig. 1 A plane-based geometric simplification of the concurrent density axis found by these coordinate transformations.

2.1 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

2.1.1 Earth's Kinematic Velocity From Electromagnetic Parameters

By solving γ for v, let us define $v_{(\gamma)}$ to be

$$v_{(\gamma)} = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{4}$$

And $\delta_{(M,R)}$ to be

$$\delta_{(M,R)} = \frac{d}{dR}g = -2\frac{GM}{R^3} \tag{5}$$

If $\delta_{(M,R)}$ is induced by spatial dilation, we should presume that this dilation is radial (at least in a simplified single body system). We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega_{(M,R)} = \sqrt[3]{\frac{d}{dR}g} = \delta_{(M,R)}^{1/3} \tag{6}$$

If we investigate this quantity for Earth, we find:

$$\omega_{\oplus} = 0.014537\ldots \tag{7}$$

While this approximates 2α, it more closely gives

$$\omega_{\oplus} = \alpha\,(2-\alpha) \pm 0.0253\% \tag{8}$$

Then by expanding equation 8

$$\alpha\,(2-\alpha) = \sqrt[3]{\frac{d}{dR}g} \tag{9}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4\left(\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6}\right) = 1.272 \times 10^{21} \tag{10}$$

If we then take the radius independent magnitude, isolating the additional electromagnetic parameters we find

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{11}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 4. This gives

$$\gamma_{\text{em}} = 1 + \frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{12}$$

$$v_{(\gamma_{\text{em}})} = 371,721\ \frac{m}{s} \tag{13}$$

matching the 2018 Plank data[12] to within $\approx 0.52\%$.

2.1.2 Solar ω

If we expand equation 9 we find

$$\alpha = \Gamma_{(\sigma)} \left(\frac{d}{dR} g \right)^{1/3} \tag{14}$$

Where $\Gamma_{(\sigma)}$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of σ.

Table 1 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

In equation 9 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{(\sigma),\oplus} \approx \frac{1}{2 - \sigma} \tag{15}$$

where $\sigma_{\oplus} \approx \alpha$.

The Sun, being the dominant local mass with a surface gravitational acceleration almost 30 times greater than the next body in question, would project this spatial vector (moving space itself) more closely to a purely radial projection. This behavior, the manner in which the Sun dominates the local gravitational environment should allow us to approximate a single body solution near the solar surface.

Let us then inspect $\Gamma_{(\sigma)}$, under the assumption that this quantity acts *inside* of the divergence, acting on each individual $\mathbb{R}^3$ axis.

$$\Gamma_{(\sigma),\odot} \propto \sqrt[3]{\Gamma_{(\theta)}} \tag{16}$$

If we then presume that σ is associated with spin, we can approximate this term as $\sigma \approx 0$, giving

$$\Gamma_{(\sigma),\odot} \approx \frac{1}{\sqrt[3]{2}} \tag{17}$$

Or

$$\alpha = \Gamma_{(\sigma),\odot} \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{18}$$

We then find through observation that for the least mass dense body in the system, $\Gamma_{(\sigma),\hbar} \approx 1$.

This gives a set of 3 equations where

$$
\begin{aligned}
g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3\,(2-\alpha)^3}{2}\right) && \pm 0.008\% \\
g_{\odot} &= R_{\odot}\alpha^3 && \pm 1.426\% \\
g_{\hbar} &= \frac{1}{2}R_{\hbar}\alpha^3 && \pm 1.134\%
\end{aligned}
\tag{19}
$$

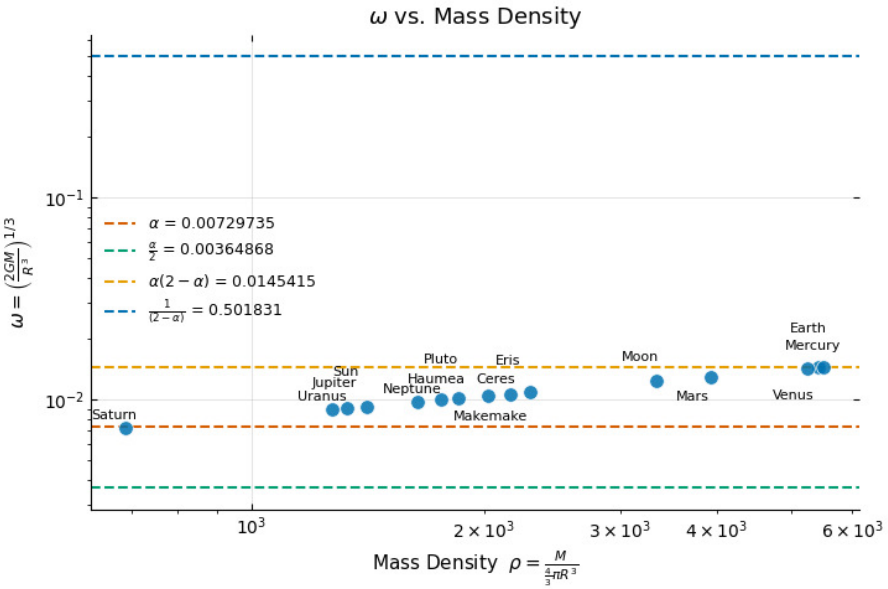


Fig. 2 The inverse of the determinant representing a change in volume, being equivalent to dy, plotted against mass density.

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha\,(2-\alpha)$ as shown in figure 2.

2.2 Gravitational Parameters

2.2.1 Cosmological Velocity

$$
\varphi_{(M,R)} = \frac{1}{R}\int \delta_{(M,R)} = 1 + G\frac{M}{R^3} \tag{20}
$$

$$
v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2_{(M,R)}}} = 525,493.17\ \text{m/s} \tag{21}
$$

2.2.2 Kinematic Velocity

$$
\int_0^R \int_0^x \left[\delta\frac{1}{v_\gamma}\tau\right]\, dR\, d\tau = 1 + \frac{1}{2}\frac{GM_{\oplus}}{R^3_{\oplus}}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{22}
$$

$$v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,567 \text{ m/s} \tag{23}$$

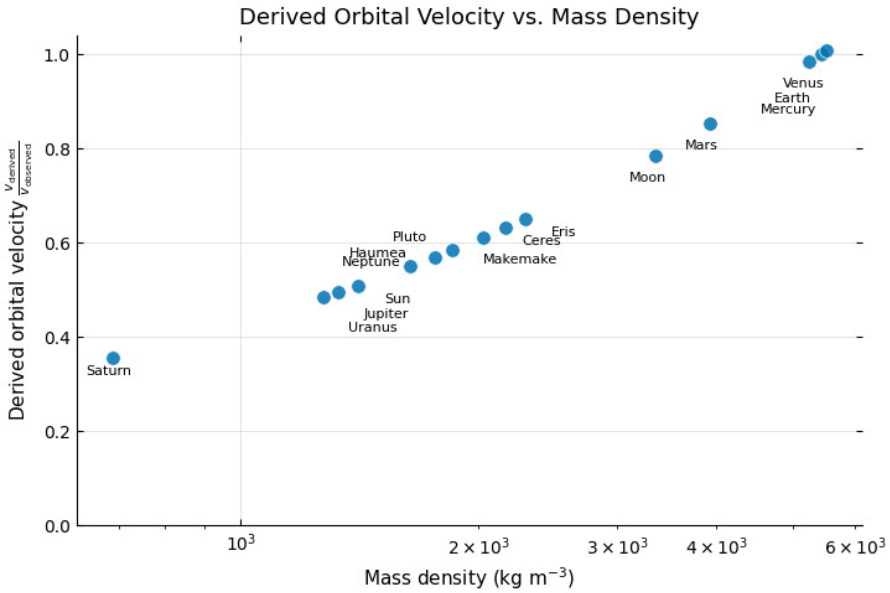


Fig. 3 A trigonometric derivation of a relativistic orbital velocity equation.

2.3 Orbital Velocity

2.3.1 Relativistic Stable Orbit Velocity

$$R_0^2 + \dot{x}^2 dt^2 = \left(\int_0^{R_0} 2G\frac{M}{R^3}\right)^2 = (R_0 + g\ dt)^2 \tag{24}$$

$$v_{\text{orbit}} = \frac{\sqrt{GM\left(GM + 2R^3\right)}}{R^2} = 230,145 \text{ m/s} = v_{\text{eilers}} \pm 0.498\% \tag{25}$$

2.3.2 Kinematic Velocity From Triginometric Orbit Equations

$$\mathrm{d}t = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{26}$$

$$v_{(-\mathrm{d}t)} = 371,580 \text{ m/s} \tag{27}$$

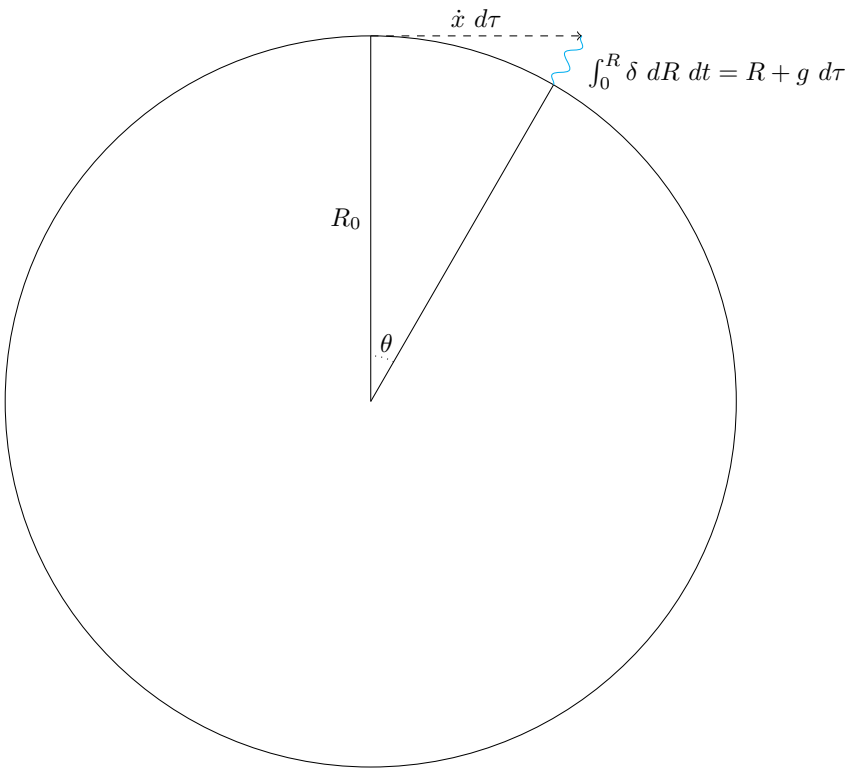


Fig. 4 A trigonometric derivation of a relativistic orbital velocity equation.

2.4 Fifth-Dimensional Spatial Displacement and Gravitational Force

2.4.1 Geometric Construction of the Fifth-Dimensional Displacement

We extend the ordinary three-dimensional spatial displacement into the additional spatial coordinate y. Let the magnitude of the ordinary spatial displacement be

$$dx = \sqrt{dx_i dx^i}, \qquad i = 1, 2, 3. \tag{28}$$

For a spatial velocity magnitude v, the displacement parameter τ defined in the theoretical framework gives

$$dx = v\tau. \tag{29}$$

The extended displacement forms the hypotenuse of a right triangle whose perpendicular components are dx and dy. Using the Lorentz factor γ already used in Equation 4, the hypotenuse is

$$dx^{(U)} = v\tau\gamma. \tag{30}$$

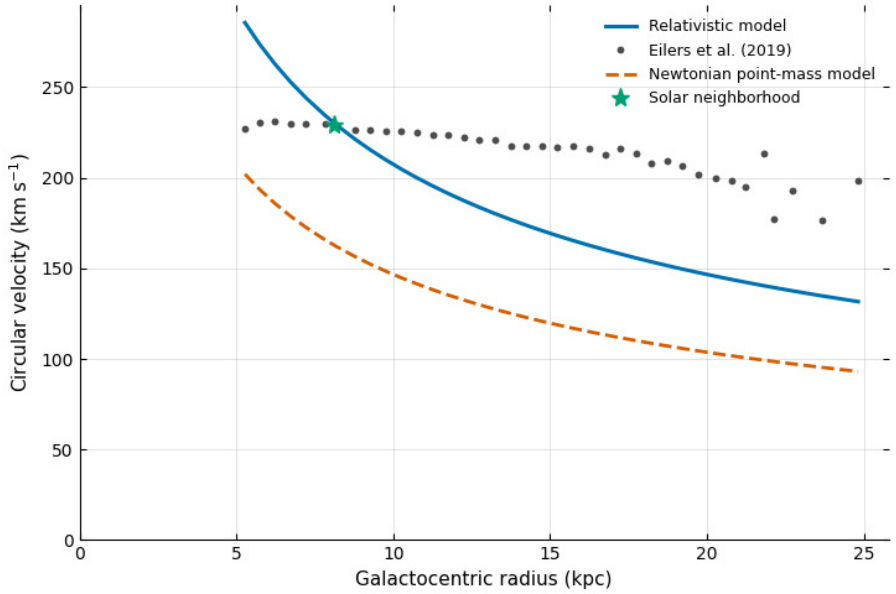


Fig. 5 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

Applying the Pythagorean theorem,

$$\left(dx^{(U)}\right)^2 = dx^2 + dy^2, \tag{31}$$

and substituting the expressions for the ordinary and extended displacements gives

$$dy^2 = v^2\tau^2\gamma^2 - v^2\tau^2 \tag{32}$$

$$= v^2\tau^2(\gamma^2 - 1). \tag{33}$$

Taking the positive magnitude of the additional spatial displacement,

$$dy = v\tau\sqrt{\gamma^2 - 1}. \tag{34}$$

Using the relation between v and γ in Equation 4, this becomes

$$\boxed{dy = \tau\frac{v^2}{\sqrt{c^2 - v^2}}} \tag{35}$$

Thus, the displacement along the additional spatial coordinate is expressed in terms of the ordinary spatial velocity magnitude and the displacement parameter.

Differentiating with respect to τ gives

$$\boxed{\frac{dy}{d\tau} = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{36}$$

2.5 Incorporation into the Kaluza–Klein Metric

We represent the additional spatial coordinate through the metric component G_{55}. For a diagonal metric, the five-dimensional line element is

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{55}dy^2. \tag{37}$$

The corresponding metric tensor has the block structure

$$\boxed{G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & G_{55} \end{pmatrix}} \tag{38}$$

Here, G_{55} describes the metric scaling of the additional spatial coordinate. Its spatial variation determines how displacement along that coordinate couples to motion in the ordinary spatial coordinates.

For a radial coordinate R, the relevant Christoffel symbol is

$$\Gamma^R_{55} = -\frac{1}{2}g^{RR}\frac{\partial G_{55}}{\partial R}. \tag{39}$$

For a normalized radial coordinate with $g^{RR} = 1$,

$$\boxed{\Gamma^R_{55} = -\frac{1}{2}\frac{\partial G_{55}}{\partial R}} \tag{40}$$

2.6 Derivation of the Radial Force Contribution

The radial geodesic equation, using τ as its trajectory parameter and assuming no angular motion, is

$$\frac{d^2R}{d\tau^2} + \Gamma^R_{55}\left(\frac{dy}{d\tau}\right)^2 = 0. \tag{41}$$

Substituting Equation 40 yields

$$\frac{d^2R}{d\tau^2} = \frac{1}{2}\frac{\partial G_{55}}{\partial R}\left(\frac{dy}{d\tau}\right)^2. \tag{42}$$

Using the fifth-dimensional displacement rate in Equation 36,

$$\frac{d^2R}{d\tau^2} = \frac{1}{2}\frac{\partial G_{55}}{\partial R}\frac{v^4}{c^2 - v^2}. \tag{43}$$

Defining the corresponding generalized radial force contribution for a body of mass m as $\mathcal{F}_R = m\,d^2R/d\tau^2$, we obtain

$$\boxed{\mathcal{F}_R = \frac{mv^4}{2(c^2 - v^2)}\frac{\partial G_{55}}{\partial R}} \tag{44}$$

This relates the gradient of the fifth-dimensional metric component to its radial force contribution. The velocity-dependent factor follows from the geometric extension of the ordinary spatial displacement, while the metric gradient sets the strength and direction of the radial coupling.

Rearranging Equation 44 gives

$$\boxed{\frac{\partial G_{55}}{\partial R} = \frac{2(c^2 - v^2)}{mv^4}\mathcal{F}_R} \tag{45}$$

Thus, a known radial force constrains the spatial gradient of G_{55}, while a specified fifth-dimensional metric determines the corresponding radial force contribution through Equation 44.

The fifth-dimensional coordinate is treated as linear and non-compact; no periodic identification or compactification condition is imposed.

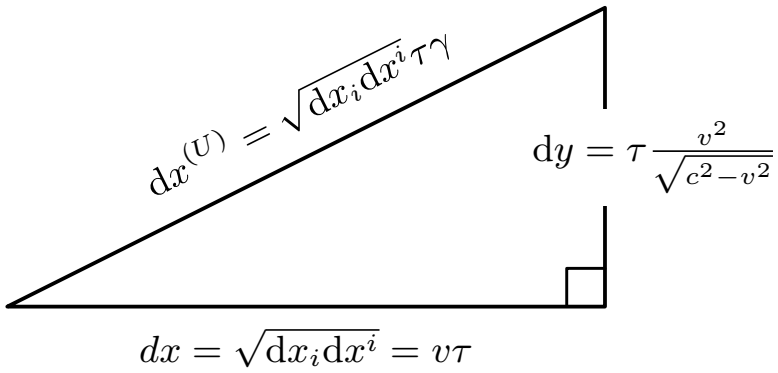


Fig. 6 Right-triangle construction used to represent the ordinary spatial displacement dx, the additional spatial displacement dy, and the extended displacement $dx^{(U)}$. The relation assigned to the hypotenuse is an assumption of the proposed geometric construction.

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